



China's Coal Buildout Signals Strategic Preparation for Taiwan Conflict

CAI 6990: Capstone Project

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Introduction

China's energy development is often talked about in the context of economic growth or environmental impact, but there is an indication of something bigger that often gets overlooked. On paper, China has committed to climate goals and a green energy transition, but in practice, they are building new coal-fired power plants at a pace that raises serious questions. This research applies an anticipatory intelligence lens to analyze the possible significance of this rapid expansion of coal-fired energy infrastructure. Looking for early warning signs and applying foresight skills, this coal expansion could indicate China is preparing for conflict.

Taiwan has become one of the most watched and arguably, most dangerous flashpoints in the Indo-Pacific. China has made its intentions of "reunification of Taiwan with the mainland" clear through military drills, cyber operations, and media statements.¹ If history can serve as a lesson, some of the strongest signals of strategic preparation are coming from much quieter sources such as energy infrastructure. Although most countries have been largely shifting away from coal-fired power, it has become an important component of China's civil-military fusion strategy because it's a reliable, controllable, and nationally sourced energy solution. If China were preparing for a war that could isolate it from foreign energy imports, coal would play a huge role in keeping its economy and military running.² By mapping where and how fast these plants are being built, looking at how they are funded, and analyzing the messaging China is releasing about them, a better understanding of China's possible underlying intentions can be developed.

1 "China Is Battening down for the Gathering Storm over Taiwan." War on the Rocks, April 17, 2024. <https://warontherocks.com/2024/04/china-is-battening-down-for-the-gathering-storm-over-taiwan/>.

2 Collins, Gabriel. "Energy Stockpiling as a China Strategic Warning Indicator." Baker Institute, 2024. https://www.bakerinstitute.org/sites/default/files/2024-06/Tes-Collins-Energy-Stockpiling-062024_0.pdf.

Scope and Problem Statement

The central question of this project: ***Is China's push to build more coal-fired power plants part of a broader strategy to prepare for a potential conflict over Taiwan?*** More specifically, is it about building energy resilience that would allow China to endure sanctions, blockades, or supply chain disruptions during a geopolitical crisis?

The focus is on coal-fired power infrastructure between 2021 and 2025. This research includes open-source data, satellite imagery, announcements from the Chinese government, and analyses from sources outside of China to track where new plants are going up and why. The research is specifically focused on coastal provinces like Fujian, Guangdong, and Zhejiang. These regions are not only important economically, they are also home to key military assets, and have close proximity to Taiwan across the strait. Layering energy infrastructure data on top of military maps gave insight into patterns and overlap. In China, the lines between civilian and military sectors are often blurred. Infrastructure decisions are not always just about efficiency, they are often about strategy. The goal of this project was not to “prove” that China is planning an invasion. Instead, it is to identify possible signals that, when combined with other components, might offer a clearer picture of what's coming.

Methodology

This project used a foresight-based methodology to analyze how China's energy and military systems interact. The analysis began by identifying key systems, including energy infrastructure, global energy trade, and military logistics, and assessing their current state. Then the scenario of a Chinese invasion of Taiwan was applied to examine how these systems might respond under stress.

The accompanying visualization aims to display the large quantities of data available using an interactive map that shows the distribution of coal-fired power plants across China. The map uses data from Global Energy Monitor (GEM), which includes information about both current operating capacity and newly permitted capacity for coal plants. This visualization was designed to help identify patterns across regions, especially in areas close to Taiwan. GEM's coal plant tracker served as the primary dataset for tracking infrastructure development but, where appropriate, satellite imagery was used to verify coal plant construction locations. Chinese state-run media sources like China Daily and South China Morning Post were also used, with the acknowledgement of government-driven framing.

China's Energy Resilience Strategy

China currently has more than 1.19 terawatts (TW) of coal-fired generation capacity, about 36% of its total installed power. Even though the country is investing heavily in renewables like solar (890 GW) and wind (520 GW), coal still supplies nearly 60% of its electricity. For context, that's about six times more coal capacity than the entire U.S. has.³

So why does coal still matter so much for China? First, it's domestically sourced; more than 70% of China's coal is mined within its borders. In 2024 alone, China hit a record 4.76 billion tonnes of production.⁴ That means even if trade routes are blocked or oil imports get cut off, coal gives China a solid backup. From a strategic point of view, this makes coal not just an energy source, but a tool of self-reliance in case of conflict.

Xi Jinping's administration has been pushing what it calls "dual circulation" which means increasing domestic production to build up resilience from economic turmoil. For the energy sector, this means establishing more ways to become self-sufficient. The idea is to keep the grid stable and the economy running no matter what.⁵ State media recently reported that China's energy self-sufficiency has now surpassed 80%, which is a major national security achievement for China.⁶

Another factor is coal's flexibility. Many of China's newer coal plants can scale up or down quickly, acting as support when solar or wind output dips. They use ultra-supercritical turbines and are connected to ultra-high voltage lines that can deliver electricity across long distances to either industrial or military sites. That flexibility is key when planning for possible instability that comes from geopolitical conflict, and the increased need for power that military operations require.⁷ An invasion of Taiwan would also likely subject China to power outages, cyberattacks, and disrupted supply chains.

If conflict were to break out over Taiwan, liquified natural gas terminals and oil refineries near the coast would be vulnerable to missile strikes or blockades. But inland coal plants could be placed strategically to be more protected from strikes and would be able to keep running and supply power to military and civilian operations. These coal-fired plants could serve as fallback systems for China.⁸

3 "China Puts Coal on Back Burner as Renewables Soar." Global Energy Monitor, August 21, 2024. <https://globalenergymonitor.org/press-release/china-hits-the-brakes-on-coal-power-permits-but-constructions-remain-robust/>.

4 IEA. "China - Countries & Regions." IEA. Accessed March 10, 2025. <https://www.iea.org/countries/china/energy-mix>.

5 "Dual Circulation to Protect China Economy in 'extreme' Circumstances: Xi." South China Morning Post, June 9, 2023. <https://www.scmp.com/news/china/politics/article/3223483/dual-circulation-needed-protect-china-economy-extreme-circumstances-xi-jinping-warns>.

6 "Energy Self-Sufficiency Pointer to Secure Future." Chinadaily.com.cn, April 1, 2025. <https://www.chinadaily.com.cn/a/202503/31/WS67e9ec83a3101d4e4dc2bbd7.html>.

7 "China Remains World Leader in Clean Coal-Fired Power Generation Technologies - the Global Energy Association." The Global Energy Association - Global Energy, March 29, 2024. <https://globalenergyprize.org/en/2024/03/29/china-remains-world-leader-in-clean-coal-fired-power-generation-technologies/>.

8 Collins, Gabriel. "Energy Stockpiling as a China Strategic Warning Indicator." Baker Institute, 2024. https://www.bakerinstitute.org/sites/default/files/2024-06/Tes-Collins-Energy-Stockpiling-062024_0.pdf.

Coal-Fired Power Plant Expansion

From 2021 to 2024, China approved more than 300 new coal-fired power units, totaling over 200 gigawatts of capacity. Construction has started on 94.5GW of new coal-fired plants in 2024, a 10-year high for the country.⁹ In 2023, they approved 114 GW of new coal power, the highest annual figure since 2015, and started building over 70 GW, more than the total coal capacity of some major economies.¹⁰ According to Global Energy Monitor, new coal plants are now being approved at a rate of about two per week.¹¹

Looking at where these plants are being built reveals even more. There's a noticeable cluster of new projects in provinces like Fujian, Zhejiang, and Guangdong, places that are not only key to China's economy but also packed with military infrastructure and sitting right across from Taiwan. Satellite imagery and open-source energy infrastructure maps show new power plants going up within 50 kilometers of major PLA bases, naval ports, and logistics hubs.¹²

Many of these coal plants are connected to China's high-voltage transmission network, which allows power to be rapidly sent from one part of the country to another, whether to keep a city online or to supply a military base. Unlike oil or gas, coal is stockpiled in massive quantities and moved by train or barge across China's domestic transport system. That makes it much more resilient in a scenario where maritime shipping routes are compromised.¹³ The speed of construction of these plants also gives them a strategic advantage. The ability to construct coal-fired power plants quickly is in line with China's broader shift towards fast and flexible mobilization.

The Central Military Commission published new guidelines emphasizing the need to transition to wartime logistics quickly, and that includes ensuring energy access.¹⁴ At the same time, China has directed provincial governments to build up emergency coal stockpiles. In Fujian and Guangdong, the goal is to have 30 days of reserve fuel ready at all times, enough to keep the grid and critical infrastructure running if trade is disrupted. Since 2020, China's crude-storage capacity has increased from 1.7bn to 2bn barrels of oil, for comparison the U.S. is around 700 million barrels.¹⁵ The U.S. also consumes roughly 25% more crude oil than China. China has also directed for emergency stock piles of other critical resources such as grain and oil.¹⁶

9 Patel, Anika. "China's Construction of New Coal-Power Plants 'reached 10-Year High' in 2024." Carbon Brief, February 13, 2025. <https://www.carbonbrief.org/chinas-construction-of-new-coal-power-plants-reached-10-year-high-in-2024/#:~:text=The%20country%20began%20building%2094.5,according%2>

10 Lempriere, Molly. "China Responsible for 95% of New Coal Power Construction in 2023, Report Says." Carbon Brief, April 11, 2024. <https://www.carbonbrief.org/china-responsible-for-95-of-new-coal-power-construction-in-2023-report-says/>.

11 "China Permits Two New Coal Power Plants per Week in 2022." Global Energy Monitor, February 2023. <https://globalenergymonitor.org/wp-content/uploads/2023/02/China-permits-two-new-coal-power-plants-per-week-in-2022.pdf>.

12 "Baker Institute China Energy Map." Baker Institute. <https://www.bakerinstitute.org/chinas-energy-infrastructure>.

13 Collins, Gabriel. "Energy Stockpiling as a China Strategic Warning Indicator." Baker Institute, 2024. https://www.bakerinstitute.org/sites/default/files/2024-06/Tes-Collins-Energy-Stockpiling-062024_0.pdf.

14 "Military and Security Developments Involving the People's Republic of China." U.S. Department of Defense, 2023. <https://media.defense.gov/2024/Dec/18/2003615520/-1/-1/0/MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2024.PDF>.

15 "Frequently Asked Questions (FAQS) - U.S. Energy Information Administration (EIA)." Frequently Asked Questions (FAQS) - U.S. Energy Information Administration (EIA). Accessed January 7, 2025. <https://www.eia.gov/tools/faqs/faq.php?id=709&t=6>.

16 Collins, Gabriel. "Energy Stockpiling as a China Strategic Warning Indicator." Baker Institute, 2024. https://www.bakerinstitute.org/sites/default/files/2024-06/Tes-Collins-Energy-Stockpiling-062024_0.pdf.

Interactive Visualization

This interactive map visualizes the spatial distribution and development status of coal-fired power plants across China, using data from the Global Energy Monitor. Plants are categorized by status: operating (green), under construction or permitted (light green), and cancelled (red), and sized according to their generating capacity in megawatts. The map also includes key PLA military installations (blue) and a reference marker for the Taiwan Strait midpoint (orange).

View the map here: [China Coal Plant Interactive Map](#)

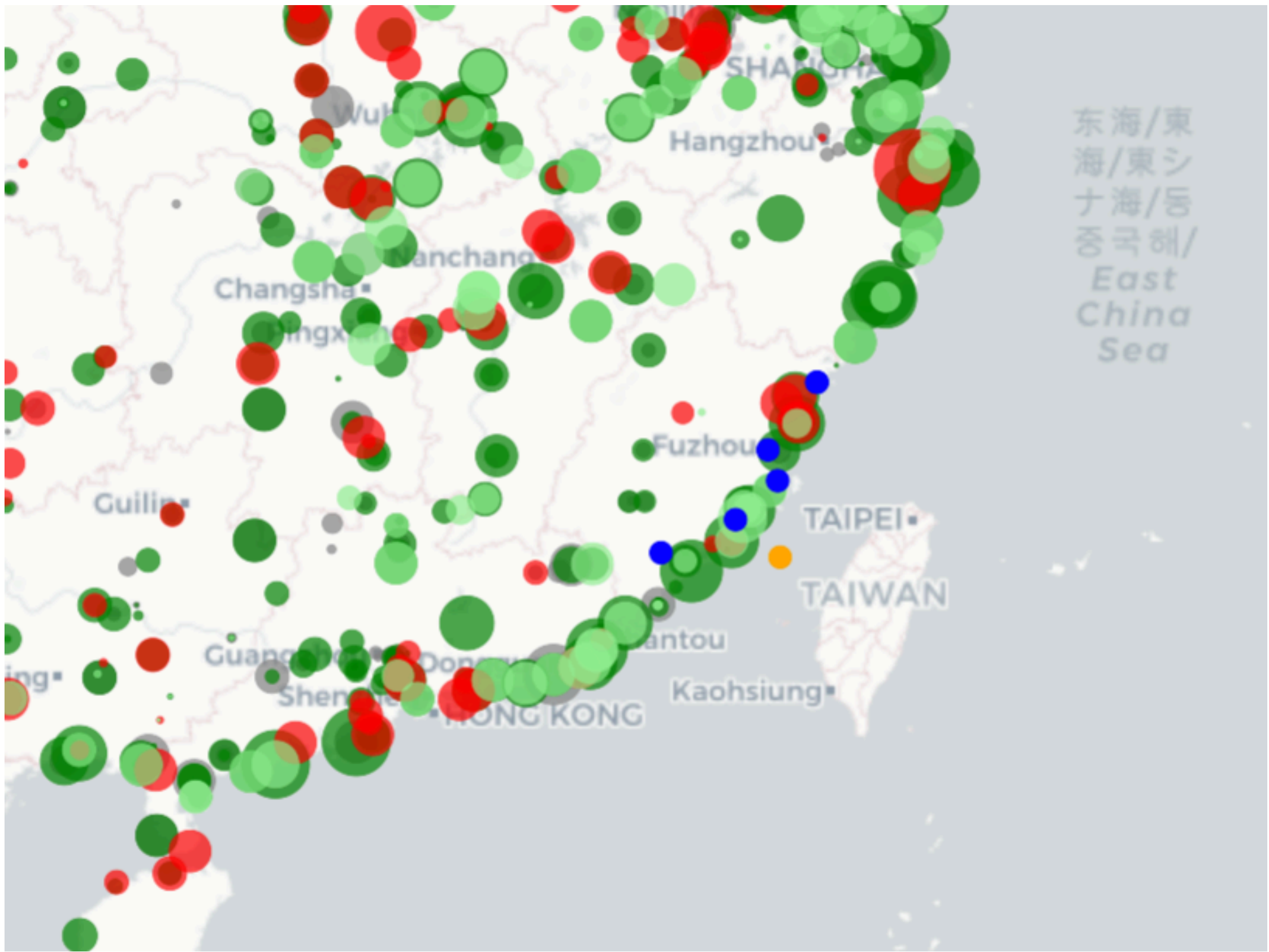


Figure 1. Screenshot of Interactive Map of Coal-Fired Power Plants in China

There is a high concentration of operating and planned coal-fired power plants in coastal provinces such as Zhejiang and Guangdong. These areas are directly across the Taiwan Strait and would play a central role in any future conflict involving Taiwan. These provinces also contain important PLA military assets, including Zhangzhou Airbase, Huian Airbase, and Longtian Airbase, all of which fall under the PLA’s Eastern Theater Command, the branch specifically tasked with operations targeting Taiwan.¹⁷ These airbases are strategically positioned to support short-range fighter jets, bombers, and surveillance aircraft capable of quickly reaching Taiwanese airspace. In recent years, these bases have been expanded or modernized to accommodate increased activity and rapid mobilization.¹⁸ Their proximity to new coal infrastructure raises the possibility that nearby energy developments are not coincidental. A steady, localized power supply would be critical to supporting high-demand operations from these bases, especially if external fuel or energy supply lines were cut.

17 S, Trishala. “Theater Commands of China.” ORCA, November 12, 2024. <https://orcasia.org/theater-commands-of-china>.

18 Blanchard, Ben. “Taiwan Warns of ‘enormous’ Chinese Bases near Its S.China Sea Holding.” Reuters, March 19, 2024. <https://www.reuters.com/world/asia-pacific/taiwan-warns-enormous-chinese-bases-near-its-schina-sea-holding-2024-03-20/>.

Indicators of Invasion Preparation

When China’s coal expansion is analyzed from all angles, it starts to look like more than just energy planning. Taken alone, each of these elements might not raise alarms. But seen together and especially considering the geopolitical background it is taking place in, they suggest a layered approach to building national resilience. China isn’t just investing in energy; it’s building resilience to endure a crisis. Although this is not definitive proof of preparation for an invasion of Taiwan, some interesting findings of this research include:

Military Proximity: New coal plants are being built near known military sites. In Fujian, several plants sit close to bases operated by the PLA’s Eastern Theater Command, which would lead any Taiwan operation.¹⁹

Grid Redundancy: These coal plants are designed to work alongside renewables not replace them. Many of them are equipped with black start capabilities, meaning they can restart power grids after a major failure or attack (a blackout).²⁰ That kind of design points to expectations of major disruptions to the rest of the grid.

Faster Timelines: The expedited construction of these plants alongside China’s recent push to improve mobilization logistics and “transition-to-war” capacity points to readiness building.²¹

Language Shift in Policy: Chinese energy documents are increasingly using terms like “energy sovereignty,” “strategic infrastructure,” and “dual-use capacity.” These mirror military planning documents and show how energy policy is being framed through a national security lens.²²

Expansion of Civil-Military Fusion Strategy: China’s leadership has pursued military integration for decades, but under Xi Jinping, it has become a formal strategy. This strategy aims to create a unified system where civilian infrastructure directly supports military readiness and national security. Energy infrastructure is part of this broader effort to optimize national resources for both economic development and defense capability.²³

Strategic Stockpiling: China’s coal reserves reached an estimated 200 million tonnes in 2024, with additional mandates in place for coastal provinces to maintain emergency supplies. This effort is part of a much larger strategy that also includes stockpiling food, crude oil, and critical tech components.²⁴

19 “Baker Institute China Energy Map.” Baker Institute. <https://www.bakerinstitute.org/chinas-energy-infrastructure>.

20 Cheng, Xue. “(PDF) Bi-Objective Optimization of Black Start Allocation for Regional Power Grid.” IOP Science, 2020. https://www.researchgate.net/publication/340735064_Bi-objective_Optimization_of_Black_Start_Allocation_for_Regional_Power_Grid.

21 Stone, Alex, and Peter Wood. “China’s Military-Civil Fusion Strategy” Air University, June 15, 2020. https://www.airuniversity.af.edu/Portals/10/CASI/documents/Research/Other-Topics/2020-06-15%20CASI_China_Military_Civil_Fusion_Strategy.pdf.

22 Stone, Alex, and Peter Wood. “China’s Military-Civil Fusion Strategy” Air University

23 Stone, Alex, and Peter Wood. “China’s Military-Civil Fusion Strategy” Air University

24 “Why Is Xi Jinping Building Secret Commodity Stockpiles?” The Economist, July 23, 2024. <https://www.economist.com/finance-and-economics/2024/07/23/why-is-xi-jinping-building-secret-commodity-stockpiles>.

Conclusion

China's coal-fired power buildout is often talked about in terms of its environmental impact. But there's another side to the story, one that matters just as much, if not more, when tensions are rising in the Taiwan Strait. The findings from this project support that the expansion of coal infrastructure looks increasingly like a deliberate strategy to harden the country against potential energy disruptions during a conflict. Coal may not be clean, but it's dependable. It's locally sourced, easy to store, and doesn't rely on vulnerable shipping routes like oil and gas do. In a war scenario, especially one where China is cut off from imports, that kind of resilience could be decisive. By building up its coal network, China isn't just powering factories or cities, it's giving itself the ability to sustain industrial production and military activity.

These developments should be treated as strategic signals. They reflect preparation, not just policy. The speed, location, and media framing of new coal plants align with broader patterns of military readiness. And if we are only watching troop movements or cyberattacks, we are going to miss these quieter, earlier, infrastructural indicators of what might come next. The more we treat infrastructure like a strategic indicator, the better chance we have at seeing major geopolitical shifts before they happen.